Yang Ouyang

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Education Experience

North Carolina State University **Aug. 2024 – Present**
Doctor of Philosophy in Computer Science *Raleigh, U.S.A*

- **Advisor:** Jung-Eun Kim
- **Research Interests:** Efficient Inference-time LLM Reasoning and Trustworthy AI
- **Award:** 2026-2027 Provost's Doctoral Fellowship

Duke University **Aug. 2022 – May 2024**
Master of Engineering in Electrical and Computer Engineering *Durham, U.S.A*

- **GPA:** 3.83 / 4.0
- **Research Experience:** Collaborated with Duke CEI on research projects in trustworthy AI.
- **Award:** Duke Kunshan University Scholarship Award (162K RMB)

Shenzhen University **Sep. 2018 – July 2022**
Bachelor of Engineering in Computer Science and Technology *Shenzhen, China*

- **GPA:** 3.75 / 4.0
- **Awards/Honor:** Two times winner of The Second Award of Studying Star in 2020 & 2021 (Ranked in 4 & 6); Outstanding Graduate of the Year 2022

Selected Publications

- [**In Proceedings of NeurIPS 2026**] **Yang Ouyang**, Jung-Eun Kim. "StepBack: Step-Level Error-Localized Resampling for Efficient Test-Time Reasoning"
- [**In Proceedings of ICML 2026**] **Yang Ouyang**, Shuhang Lin, Jung-Eun Kim. "DenseSteer: Steering Small Language Models towards Dense Math Reasoning"
- [**In Proceedings of NAACL 2025**] **Yang Ouyang**, Hengrui Gu, Shuhang Lin, Wenyue Hua, Jie Peng, Bhavya Kailkhura, Meijun Gao, Tianlong Chen, Kaixiong Zhou. "Layer-Level Self-Exposure and Patch: Affirmative Token Mitigation for Jailbreak Attack Defense"
- [**In Proceedings of ICLR 2025**] Jingyang Zhang, Jingwei Sun, Eric Yeats, **Yang Ouyang**, Martin Kuo, Jianyi Zhang, Hao Frank Yang, Hai Li. "Min-K%++: Improved Baseline for Detecting Pre-Training Data of LLMs"

Research Experience

StepBack: Step-Level Error-Localized Resampling for Efficient LLM Reasoning **Jan. 2026 – May. 2026**
In Proceedings of NeurIPS 2026 *Raleigh, U.S.A*

- Proposed **StepBack**, a training-free inference-time decoding method that improves self-consistency by localizing likely first-error positions in chain-of-thought trajectories and resampling only uncertain suffixes instead of regenerating full reasoning paths.
- Designed lightweight step-level rollback signals using **NLL increases** and **process reward model score drops** to allocate test-time compute more efficiently within reasoning trajectories.
- Evaluated across math reasoning and multi-hop QA benchmarks; the PRM-based variant improved accuracy on **10 / 14** model–dataset pairs while reducing tokens in all settings, with up to **39.8%** token savings over **SC@32**.

Large-Scale Elastic Continued Pretraining for MoE Models **July 2026 – Sep. 2026**
Project *Shanghai, China*

- Adapted elastic training to a closed-source **15B-A2B MoE model** using **1,024 GPUs**, enabling three nested, weight-sharing subnetworks (3B, 9B, 15B) for different resource budgets.
- Explored five architectural dimensions (MoE depth, expert width, expert count, attention heads, and top-k routing) through budget-constrained sampling and **importance-guided pruning**, constructing nested subnetworks under joint total-parameter and active-parameter constraints.
- Optimized shared-weight continued pretraining across **11 benchmarks**, outperforming size- and training-budget-matched standalone models by **1.6 and 3.9 points** for small and large subnetworks, respectively.

DenseSteer: Steering Small Language Models towards Dense Math Reasoning **Aug. 2025 – Jan. 2026**
In Proceedings of ICML 2026 *Raleigh, U.S.A*

- Proposed **Dense Reasoning** as a structural property of strong mathematical reasoning, empirically showing that higher reasoning density (fewer steps, more tokens per step) correlates with higher accuracy on GSM8K and OOD benchmarks.
- Developed a **training-free, inference-time activation steering** framework using only **50 rewrite-based contrastive pairs**, avoiding distribution shift and preserving low token-level NLL.
- Achieved up to **+5% GSM8K**, **+17.5% AMC**, and **+10% AIME** accuracy gains on **Qwen-2.5-3B-Instruct** and **LLaMA-3.2-3B-Instruct** comparing with heavy training-based distillation methods.

Internship Experience

Amazon Web Services | ***AI Infrastructure*** **July 2022 – Oct. 2022**

Software Engineer Intern, DeepJavaLibrary Department *Mountain View, U.S.A (remote)*

- Developed core integration components between **DJL-Serving** and **KServe**, implementing 3 **Java**-based APIs for health monitoring, model introspection, and inference data processing within the model serving stack.
- Supported **cloud-native model serving** on **Kubernetes** by configuring deployment manifests, service endpoints, and runtime parameters for containerized inference workloads.

Technical Skills

- Programming Languages: Python, Java, C++, Javascript
- Deep Learning Frameworks: PyTorch, Huggingface, Large-Scale Continued Pretraining for MoE LLM
- AI-assisted Development Tools: Cursor, Codex, Claude Code